

Multivitamin Use and the Risk of Mortality and Cancer Incidence



Although multivitamin/mineral supplements are commonly used in the United States, the efficacy of these supplements in preventing chronic disease or premature death is unclear. To assess the relation of multivitamin use with mortality and cancer, the authors prospectively examined these associations among 182,099 participants enrolled in the Multiethnic Cohort Study between 1993 and 1996 in Hawaii and California. During an average 11 years of follow-up, 28,851 deaths were identified. In Cox proportional hazards models controlling for tobacco use and other potential confounders, no associations were found between multivitamin use and mortality from all causes (for users vs. nonusers: hazard ratio = 1.07, 95% confidence interval: 0.96, 1.19 for men; hazard ratio = 0.96, 95% confidence interval: 0.85, 1.09 for women), cardiovascular diseases, or cancer. The findings did not vary across subgroups by ethnicity, age, body mass index, preexisting illness, single vitamin/mineral supplement use, hormone replacement therapy use, and smoking status. There also was no evidence indicating that multivitamin use was associated with risk of cancer, overall or at major sites, such as lung, colorectum, prostate, and breast. In conclusion, there was no clear decrease or increase in mortality from all causes, cardiovascular disease, or cancer and in morbidity from overall or major cancers among multivitamin supplement users.